



Extreme Heatwave Early warning and Human Stress Index

- Problem Statement ID – 26083
- Problem Statement Title-

Extreme Heatwave Early Warning and Human

Thermal Stress Index

- Theme- Disaster Management
- PS Category- Software
- Team ID-
- Team Name - The Tesseract



Proposed Solution & Addressing the problem

- **Predict Heat-Health Risk using ML:**

We use a regression model trained on meteorological, thermal-stress, demographic and health-impact datasets to predict continuous target values such as heat-related mortality risk and hospitalization spikes. The model is evaluated on test data to measure its predictive performance.

- **Convert Predictions into Actionable GIS Risk Levels:**

A classification/risk-threshold layer converts the predicted values and relevant heat-stress parameters such as Heat Index, WBGT/UTCI and wind speed into predefined risk levels — **Low** → **Moderate** → **High** → **Extreme** — and displays them as colour-coded zones on the GIS dashboard.

- **Connect Predictions, GIS & Early-Warning Alerts:**

A backend orchestration layer processes and delivers the model outputs to the GIS dashboard and alert system. When the predicted risk reaches a predefined threshold, the system can generate location-specific early warnings, helping authorities identify high-risk areas and take timely preventive action.



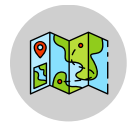
01 | Multi-Parameter Heat Monitoring



02 | Human Thermal Stress Assessment



03 | Vulnerability-Aware Risk Prediction

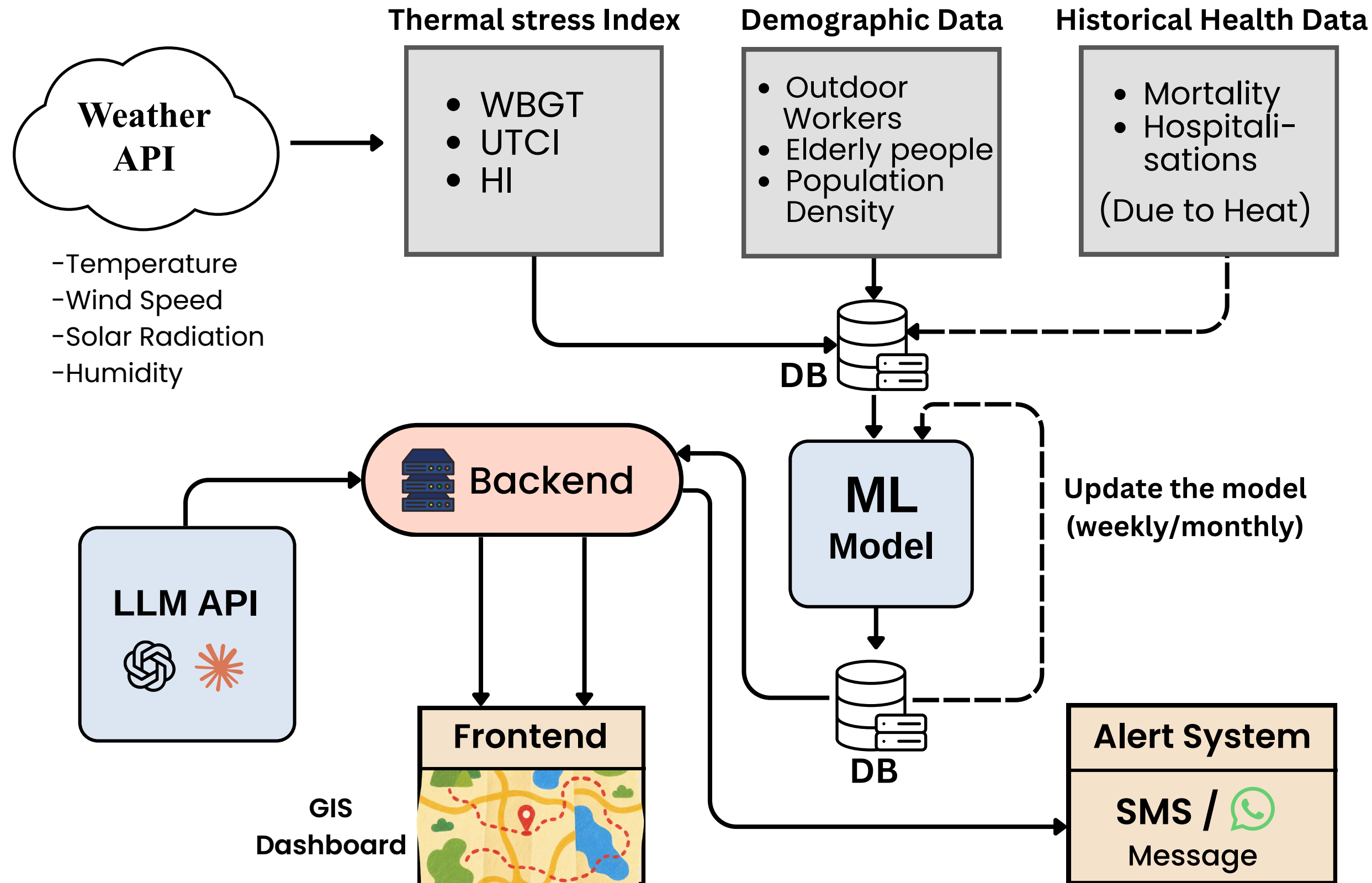


04 | Localized Early Warning & Decision Support

Innovations

- LLM API for **Advisory system** in GIS Map dashboard
- Integrated **GPS tracker** which allows system to automatically find the location of users.
- **Multilingual support** - Giving Authorities the preference for their language
- Our model can be trained to predict whenever **El Niño** affects our environment using its parameters or indicators (Niño 3.4, SSTA and SOI)





Technology used

Machine learning:

- Python 
- Scikit learn 
- Pandas 
- Numpy 

Database:

- PostgreSQL 
- PostGIS 

Frontend:

- Javascript 
- leaflet 
- Maplibre 
- OpenStreetMap 
- Twilio API 

Backend:

- Python 
- FastAPI 

Technical Feasibility

- Uses readily available weather data.
- Uses scientifically established heat-stress indices (Weather + Thermal Stress + Demographics + Historical Health Data)
- Combines multiple types of data
- Uses established AI/ML technologies
- GIS makes the prediction location-specific
- No specialized hardware is mandatory

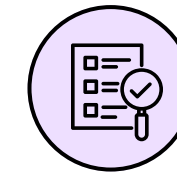
Economic Viability

- Open-source technology reduces software licensing cost
- Instead of a conventional approach with expensive hardware setup, Existing weather data + computational models + GIS significantly reduces the initial infrastructure and maintenance cost.
- But its not totally "Zero cost" where Weather API could be exceeded of its free plan, where SMS/Alert system charges are made, and for other maintenance.

Scalability

- Our model can also incorporates different vulnerable groups , not only elderly and outdoor workers.
- It can produce multiple risk warning, Not just limiting it to a single ward or district.
- The prototype will be trained and validated using synthetic data initially, with the architecture designed to incorporate validated historical health data for real-world deployment.

Potential Challenges



Data Availability & Quality



Integrate reliable weather sources and progressively incorporate validated health datasets.



Accuracy of heat stress Indices



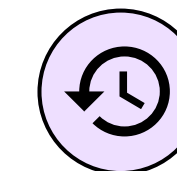
Validate WBGT/UTCI calculations and compare multiple ML models using cross-validation.



Limited Health-Impact Data



To Initially use synthetic datasets for model development and progressively validate with reliable historical hospitalization and mortality data.



Real-Time Data Processing



Automate weather-data ingestion, index calculation, prediction, and alert generation.



False/ Delayed alerts



Apply calibrated thresholds and multi-level risk classification to minimize false or delayed alerts.

1. Reduced Health Risks:

Helps prevent heatstroke, dehydration and other heat-related illnesses.

2. Protects Vulnerable Groups:

Provides targeted warnings for elderly people, children and outdoor workers.

3. Better Emergency Preparedness:

Gives authorities time to prepare healthcare, cooling centres and emergency resources.



1. Early Warning:

Predicts extreme heat conditions 3–5 days in advance for timely action.

2. Hyperlocal Alerts:

Identifies high-risk areas and provides location-specific heat stress warnings.

3. Smart Resource Allocation:

Helps authorities efficiently deploy water, medical support and cooling facilities.

Articles & Reference papers:

- https://en.wikipedia.org/wiki/Wet-bulb_globe_temperature
- <https://www.sciencedirect.com/science/article/pii/S2210670726002222>
- <https://www.frontiersin.org/journals/environmental-health/articles/10.3389/fenvh.2026.1789071/full>
- <https://www.indiatoday.in/science/story/india-heatwave-deaths-study-extreme-heat-public-health-emergency-climate-change-2920200-2026-06-01>

Geo-graphical references:

- <https://www.weather.gov/tsa/wbgt>
- https://dss.imd.gov.in/dwr_img/GIS/heatwave.html

Data sets:

- <https://developers.google.com/earth-engine/datasets/>
- <https://www.openstreetmap.org/#map>
- <https://data.who.int/>
- https://mausam.imd.gov.in/index_en.php
- <https://zenodo.org/records/15126391>